DEVOPS FINAL CAPSTONE

IronPass

From code to a monitored Kubernetes cluster

A gym membership app by Romit

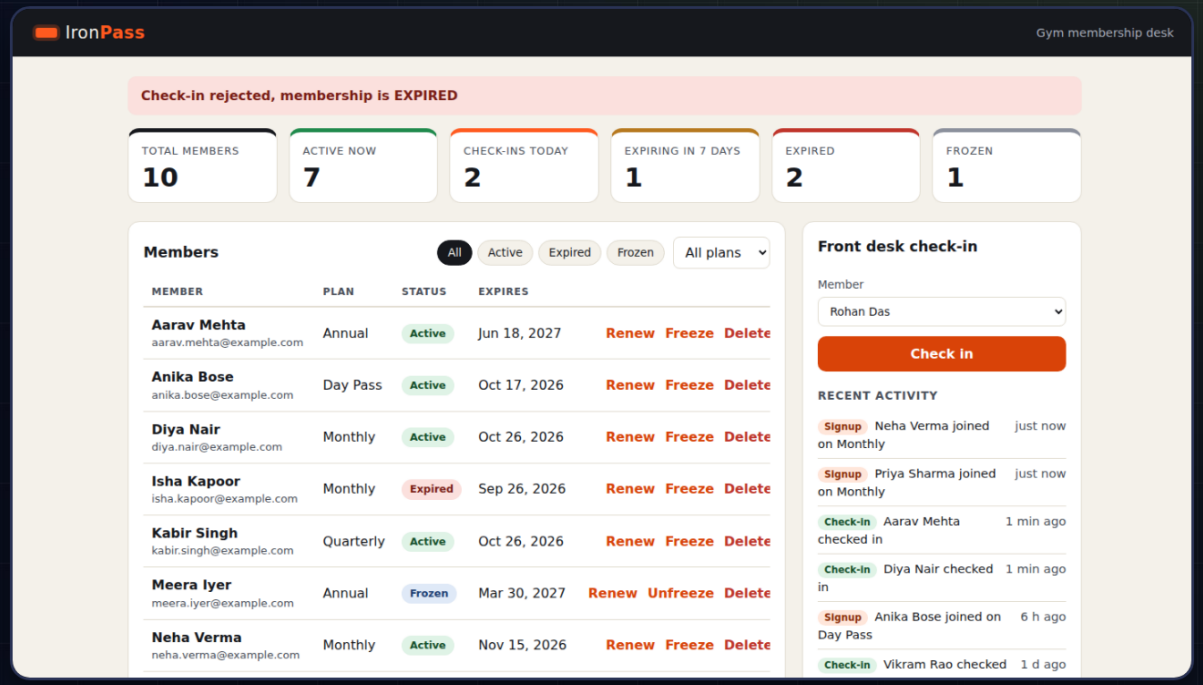


01 THE PRODUCT

The problem

Paper lists let expired members in.

- Sign members up
- Check them in
- Refuse expired plans



Expired plan, check-in refused

02 TECH STACK

Tech stack

Python backend, React frontend, PostgreSQL database, all in containers.


Python


FastAPI


React


PostgreSQL


Docker


Kubernetes



Helm


Terraform


AWS


GitHub Actions


Prometheus


Grafana

03 ARCHITECTURE

Architecture



Traefik routes web and API traffic into the cluster.

04 RUN LOCALLY

Run it with Docker

One command starts the database, the API and the website.

ironpass-devops-capstone

```
$ docker compose ps --format 'table {{.Service}}\t{{.Status}}'
SERVICE    STATUS
backend    Up 12 seconds (healthy)
frontend   Up 7 seconds (healthy)
postgres   Up 17 seconds (healthy)
```

3

services, in order

3 of 3

clean starts healthy

0

records lost on restart

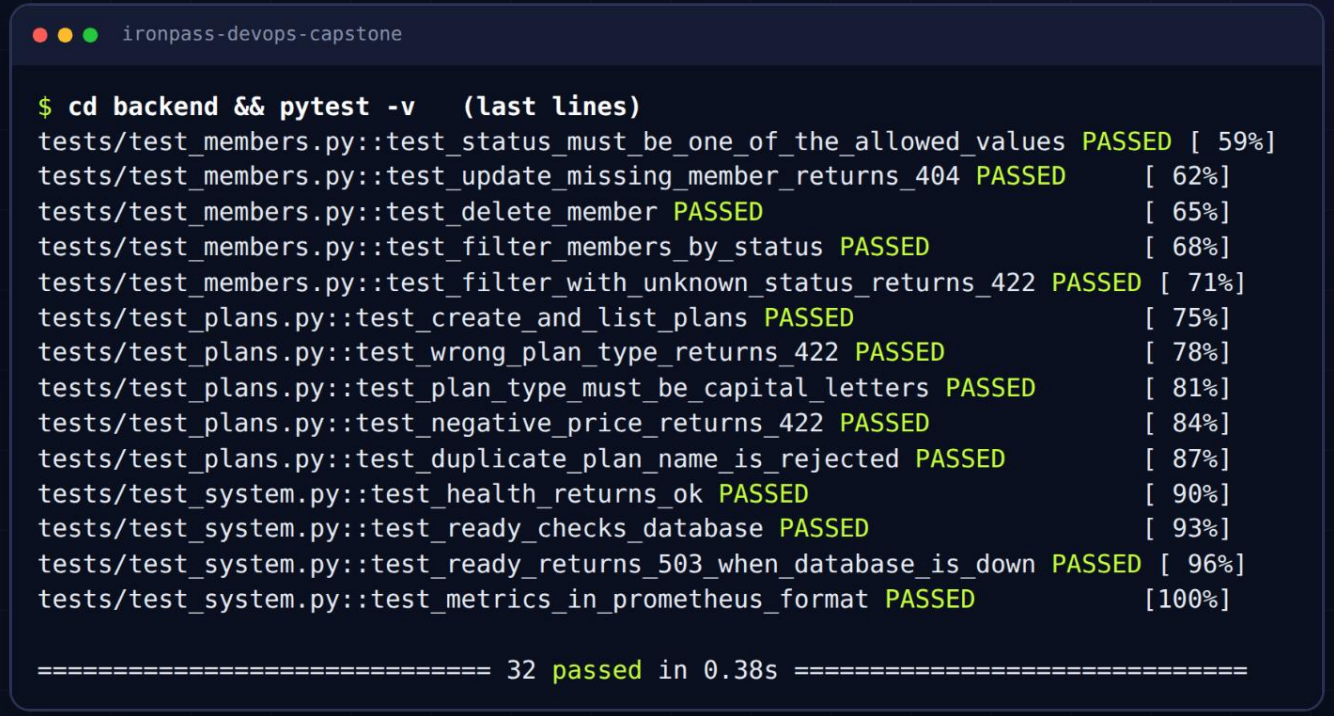
05 TEST

Tests

32

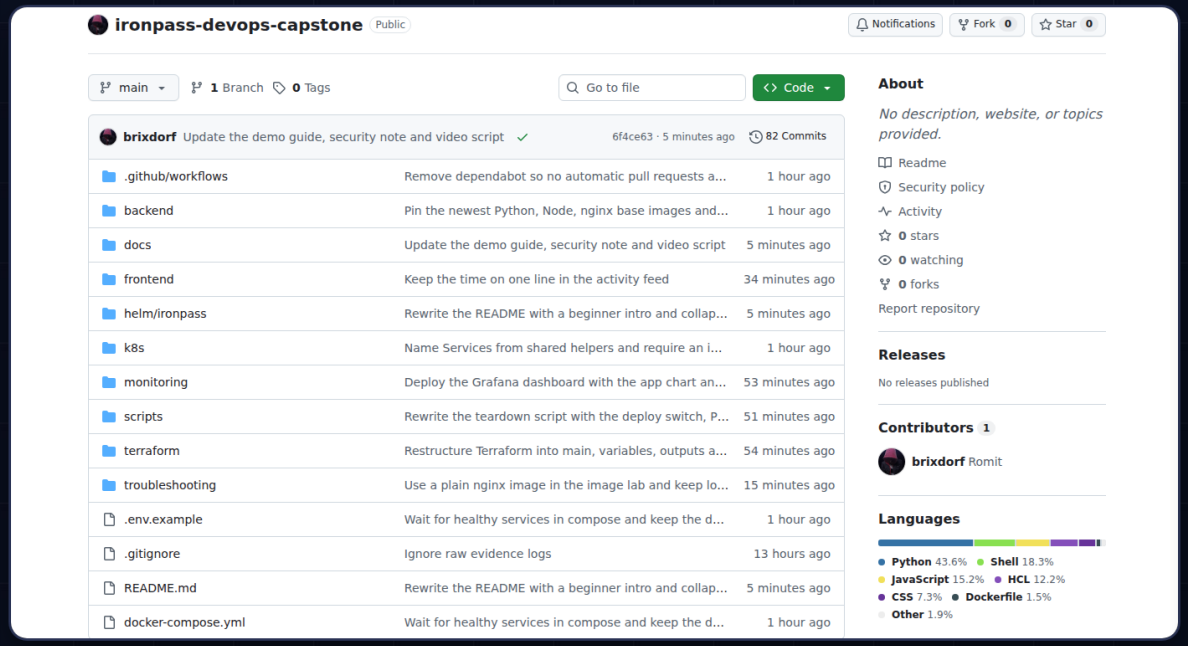
tests pass

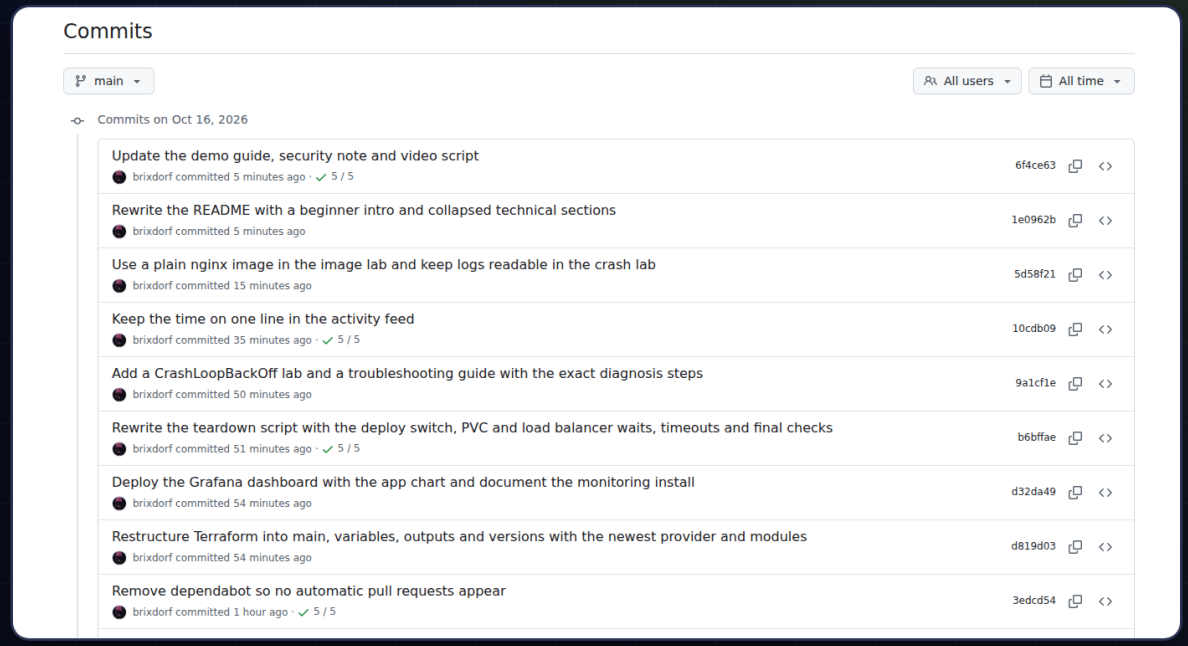
Checks wrong input, missing members and the expired plan rule.



06 CODE

Git and GitHub





85+ commits

Small commits, clear messages, no secrets in git.

07 CI/CD

The pipeline

Code
git push

Test
pytest, build

Build
one image each

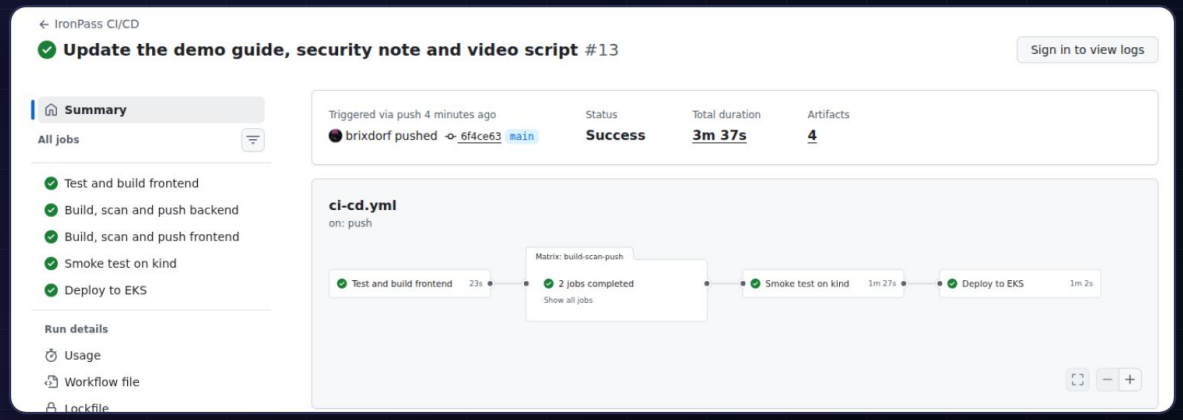
Scan
Trivy gate

Push
GHCR, SHA tag

Deploy
kind, then EKS

Monitor
Prometheus, Grafana

Provision Terraform builds the VPC, the EKS cluster and the GitHub sign-in before the first deploy



Tests gate everything

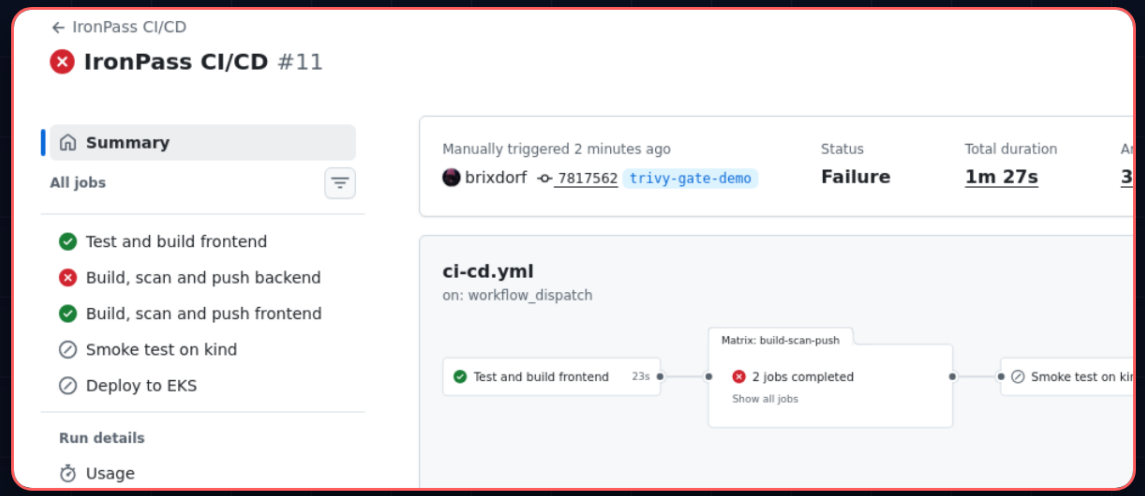
Scanned image is pushed

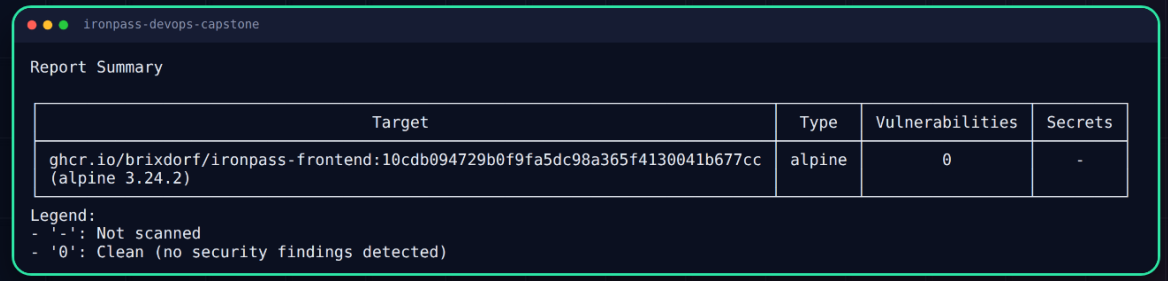
Smoke test, then EKS

08 SCAN

Security scan

Trivy fails the build on fixable HIGH or CRITICAL issues.



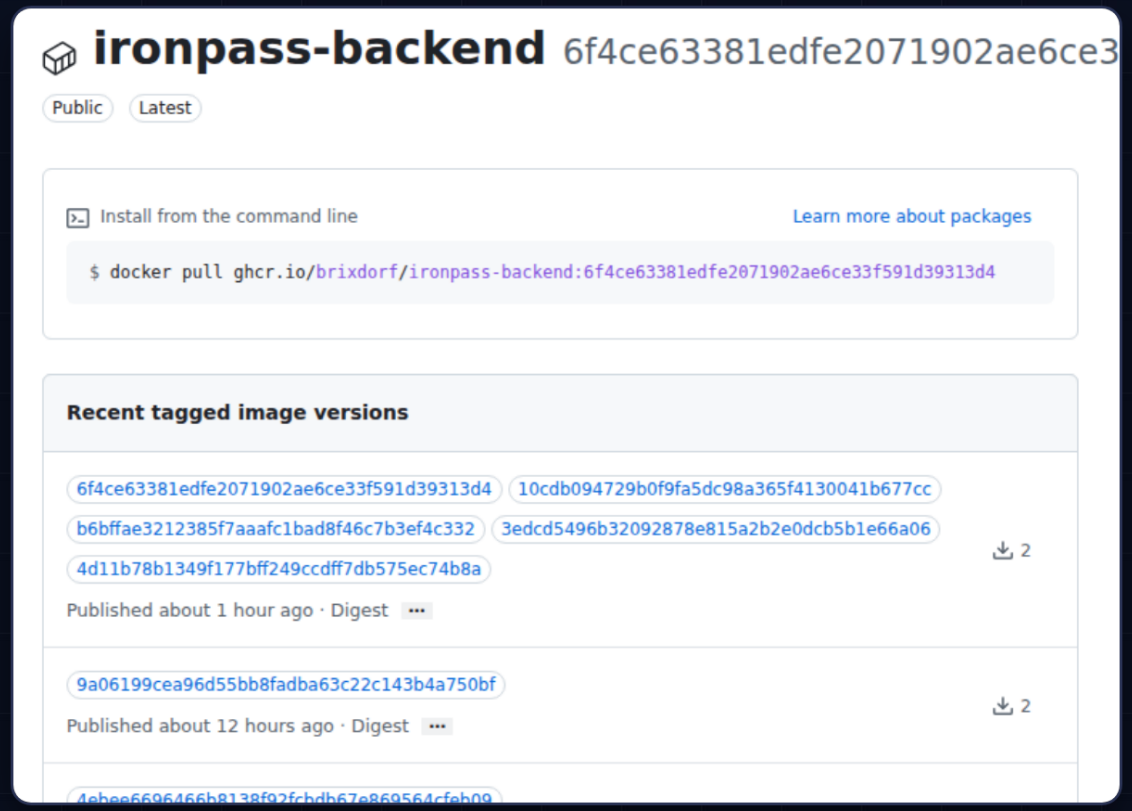


A bad image stops the pipeline

My real images: 0 findings

09 REGISTRY

Container registry



- Tagged by commit
- Built-in GITHUB_TOKEN
- Public packages

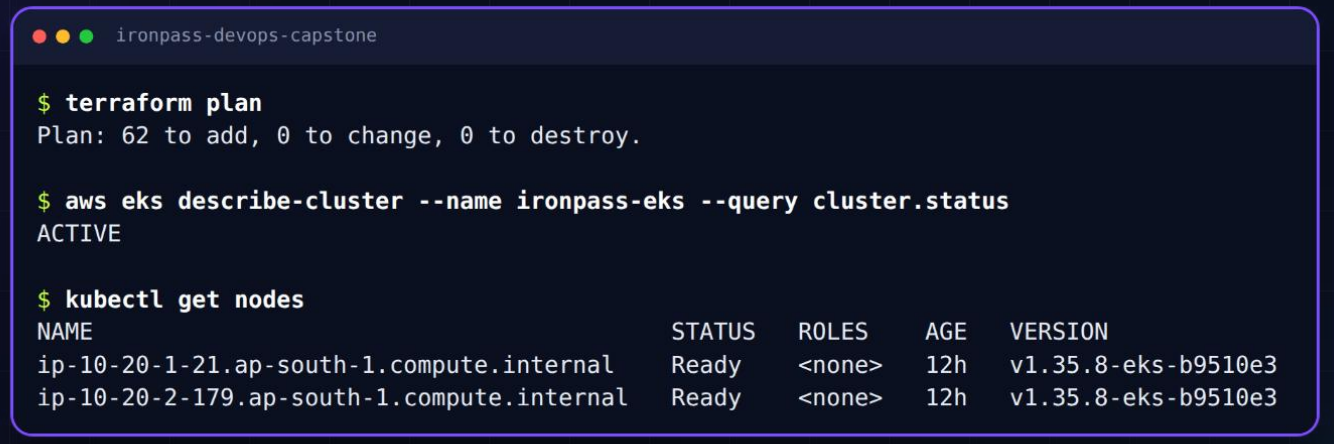
10 PROVISION

Terraform on AWS

62

resources from one plan

Code builds the network, the cluster and the GitHub sign-in.



11 DEPLOY

Kubernetes and Helm

ironpass-devops-capstone

```
$ kubectl get pods -n ironpass
NAME                                READY   STATUS    RESTARTS   AGE
ironpass-backend-84b7f64ffb-8dqmc   1/1     Running   0          16m
ironpass-backend-84b7f64ffb-pzlkb   1/1     Running   0          16m
ironpass-frontend-6cb5cfd8ff-5fqtq  1/1     Running   0          18m
ironpass-frontend-6cb5cfd8ff-jxql9  1/1     Running   0          18m
ironpass-postgres-667869ddc4-g965t  1/1     Running   0          82m

$ kubectl get svc -n ironpass
NAME               TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
ironpass-backend   ClusterIP   172.20.58.145   <none>        8000/TCP   12h
ironpass-frontend  ClusterIP   172.20.205.171  <none>        80/TCP     12h
ironpass-postgres  ClusterIP   172.20.112.74   <none>        5432/TCP   12h

$ helm status ironpass -n ironpass | head -5
NAME: ironpass
LAST DEPLOYED: Fri Oct 16 06:27:03 2026
NAMESPACE: ironpass
STATUS: deployed
REVISION: 13
```

- 5 pods, 3 services
- One Helm chart
- Rollback in one command

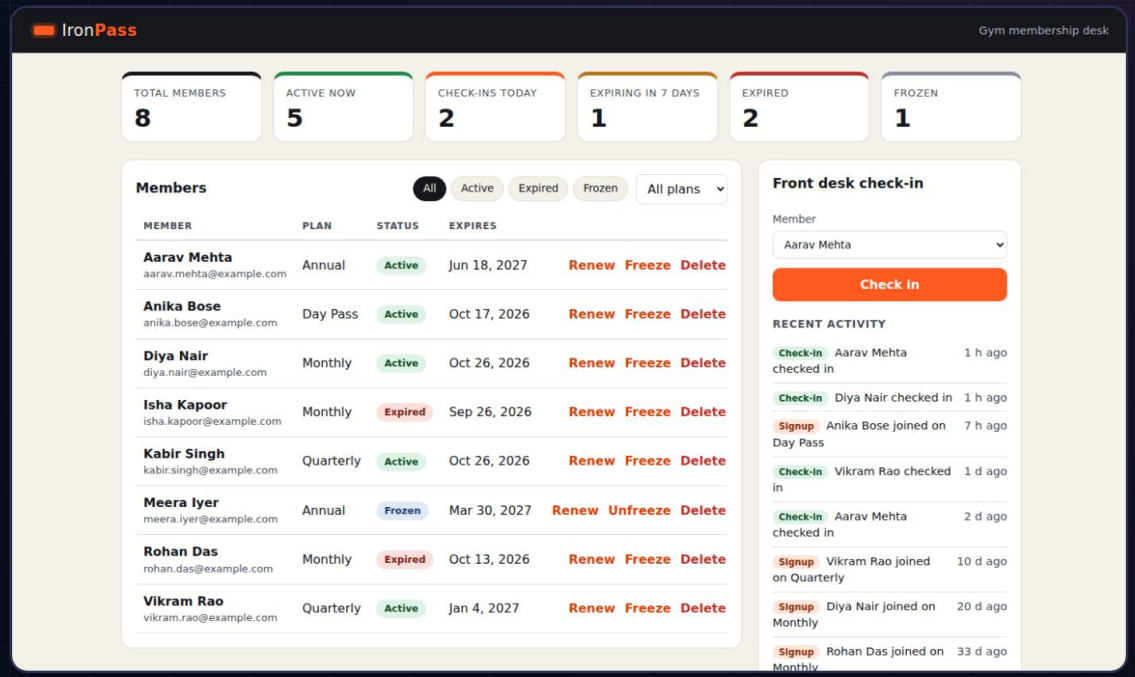
12 DEPLOY

Ingress



One address for the whole app.

Traefik replaced ingress-nginx, which is retired.



13 DEPLOY

Autoscaling

2 to 6

backend pods under load

The load script raises CPU. The autoscaler adds pods.

Before the load

ironpass-devops-capstone

```
$ kubectl get hpa -n ironpass -w   (before the load)
NAME               REFERENCE                     TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
ironpass-backend   Deployment/ironpass-backend   cpu: 2%/50%   2         6         2          12h
```

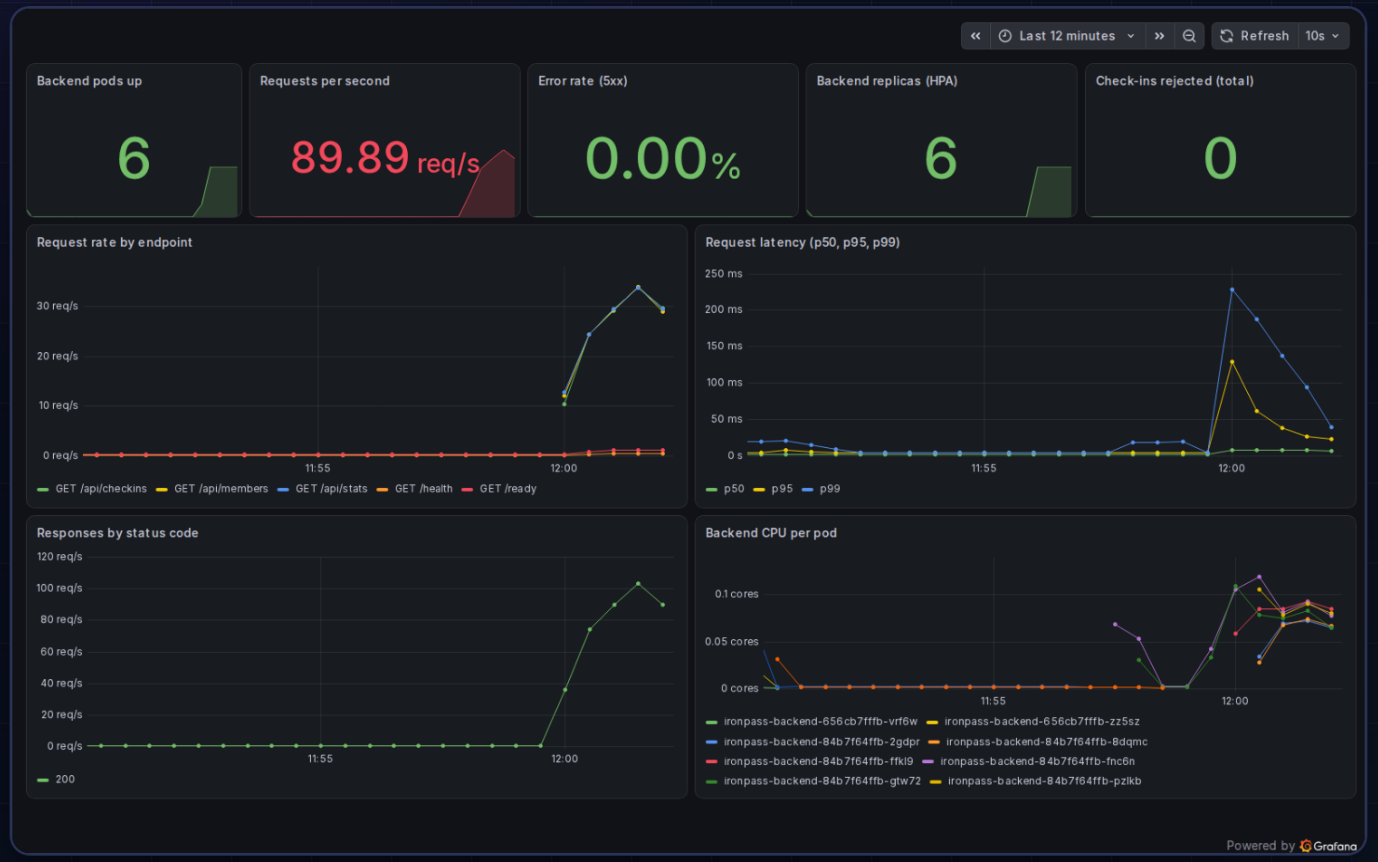
During the load

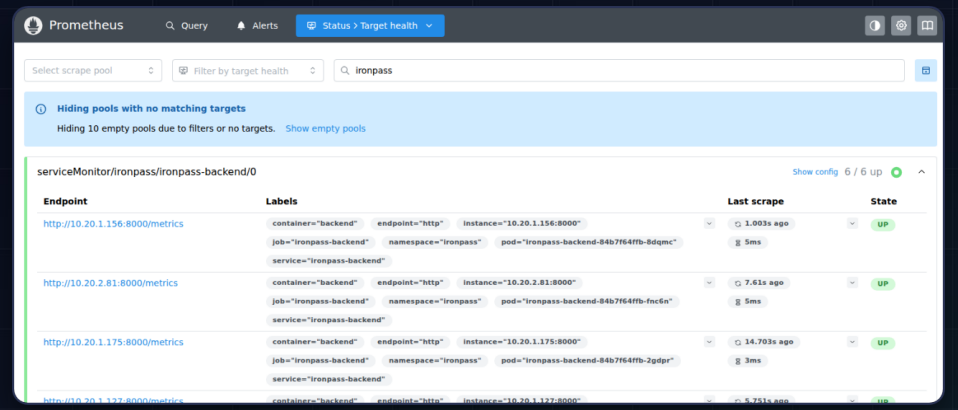
ironpass-devops-capstone

```
$ kubectl get hpa -n ironpass -w   (during the load)
NAME               REFERENCE                     TARGETS        MINPODS   MAXPODS   REPLICAS   AGE
ironpass-backend   Deployment/ironpass-backend   cpu: 87%/50%   2         6         6          12h
ironpass-backend   Deployment/ironpass-backend   cpu: 92%/50%   2         6         6          12h
```

14 MONITOR

Monitoring





Prometheus scrapes the backend. Grafana shows rate, latency and errors.

15 PRACTICE

Troubleshooting



ImagePullBackOff

Wrong image tag.



No endpoints

Label typo.



CrashLoopBackOff

Missing setting.

Found with kubectl, fixed with apply -f.

Why Kubernetes?

	Self-heals	Auto scales	Rolling updates	Any cloud
Docker Compose only	Partly	No	No	Yes
A single VM	No	No	No	Yes
Managed container service	Yes	Yes	Yes	No
Kubernetes	**Yes**	**Yes**	**Yes**	**Yes**

Kubernetes does it all, on any cloud. The price is more to learn.

17 RESULTS

Results in numbers

32 tests pass	**15** API endpoints	**3m 37s** full pipeline	**0** HIGH or CRITICAL
5 pods running	**2 to 6** backend pods (HPA)	**309 MB** backend image	**90.5 MB** frontend image

18 LEARNED

What I learned

- One job per tool
- Pin versions, scan images
- Small commits, easy fixes
- Break it, then fix it



Seven stages, one pipeline.

Thank you

From code to a monitored cluster.

github.com/brixdorf/ironpass-devops-capstone



Scan to open the repo